

PenuX at Hadassah Ein Kerem ICU ? 1-Page Collaboration Brief

Objective

Validate PenuX (early prediction of sepsis and AKI in ICU patients) on an Israeli real-world cohort at Hadas

Why Hadassah ICU

- ? Strong critical-care clinical infrastructure.
- ? Diverse internal-medicine and ICU case mix.
- ? High translational-research potential from retrospective validation to prospective workflow testing.

Project Snapshot

- ? Model scope: ICU deterioration risk (sepsis + AKI).
- ? Data foundation: MIMIC-III/IV time-series variables.
- ? Architecture: Bi-LSTM for sequential clinical data.
- ? Development approach: IEC 62304-oriented lifecycle documentation and traceability.

Proposed Validation Plan (Phase 1)

1. Retrospective cohort definition (adult ICU admissions, inclusion/exclusion criteria, target labels).
2. Data harmonization between local variables and model inputs.
3. External validation (AUROC/AUPRC, calibration, subgroup analysis).
4. Clinical utility assessment (lead time, alert burden, workflow fit).
5. Feasibility report for prospective silent-mode pilot.

What We Need from Site Collaboration

- ? Clinical PI and ICU data-science counterpart.
- ? Access pathway for de-identified retrospective ICU data (per IRB/privacy rules).
- ? Periodic clinical review meetings for outcome interpretation.

Potential Outputs

- ? Joint abstract/manuscript.
- ? Grant-ready preliminary results.
- ? Decision framework for prospective deployment pilot.

Additional Parallel Research (Optional)

- ? NEURO-LINK: EEG biomarkers for SCZ/ASD (OpenNeuro pipelines).
- ? In silico drug R&D: RDKit + VAE cohort generation with ICH E6(R2)-aligned protocol logic.

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